(c) $\sum_{n=1}^{\infty} \frac{1}{n}$ $\lim_{n\to\infty} \frac{1}{n} = 0 \leftarrow$ TFD is inconclusive.

(d) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ $\lim_{n\to\infty} \frac{1}{n^2} = 0 \leftarrow$ TFD is inconclusive.

2 Geometric Series

In a geometric series, there is a common ration between any two adjacent terms.
The common ratio is called r and the first term is a.

2.0.1 Example:

Find a and r for each series:

(a) $1 + 2 + 4 + 8 + \cdots$ Diverge

$a = 1$ $r = 2$

(b) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ Converge $\text{sum} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

$a = \frac{1}{2}$ $r = \frac{1}{2}$

(c) $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$ Converge $\text{sum} = \frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$

$a = 1$ $r = \frac{1}{3}$

(d) $12 - 3 + \frac{3}{4} - \frac{3}{16} + \cdots$ $a = 12$ $r = -\frac{1}{4}$ Converge $\text{sum} = \frac{12}{1-(-\frac{1}{4})}$

$= \frac{12}{\frac{5}{4}} = \frac{12 \cdot 4}{5} = \frac{48}{5}$

Geometric Series

$$\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$$

- A geometric series converges if and only if $|r| < 1$.
- If it converges, the sum is $\dfrac{a}{1-r}$

2.0.2 Example:

Determine a and r, then find the sum or tell if the series diverges:

(a) $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \cdots$ $= \frac{2}{1-\frac{1}{3}} = \frac{2}{\frac{2}{3}} = 2 \cdot \frac{3}{2} = 3$

$a = 2 \quad r = \frac{1}{3} \; C$

(b) $\displaystyle\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n$ $= 1 + \frac{3}{4} + \frac{9}{16} + \frac{27}{64} + \cdots = \frac{1}{1-\frac{3}{4}} = \frac{1}{\frac{1}{4}} = 4$

$a = 1 \quad r = \frac{3}{4}$

(c) $\displaystyle\sum_{n=0}^{\infty} \frac{4^{n+3}}{3^n}$ $= \overset{0}{\frac{4^3}{3^0}} + \overset{1}{\frac{4^4}{3^1}} + \overset{2}{\frac{4^5}{3^2}} + \overset{3}{\frac{4^6}{3^3}} + \cdots = \cancel{\frac{64}{1-\frac{4}{3}}} \; \infty$

$a = 64 \quad r = \frac{4}{3} \leftarrow > 1 \quad \therefore$ Diverge to ∞

(d) $\displaystyle\sum_{n=2}^{\infty} 5\left(-\frac{1}{2}\right)^n$ $\quad a = 5\left(-\frac{1}{2}\right)^2 = 5\left(\frac{1}{4}\right) = \frac{5}{4} \qquad r = -\frac{1}{2} \; C$

$\text{sum} = \frac{\frac{5}{4}}{1-\left(-\frac{1}{2}\right)} = \frac{\frac{5}{4}}{\frac{3}{2}} = \frac{5}{4} \cdot \frac{2}{3} = \frac{5}{6}$

(e) $\displaystyle\sum_{n=0}^{\infty} \frac{8^n}{3^{2n}}$ $= \overset{0}{1} + \overset{1}{\frac{8}{3^2}} + \overset{2}{\frac{8^2}{3^4}} + \cdots \qquad r = \frac{8}{9} \quad a = 1$

$= \displaystyle\sum_{n=0}^{\infty} \frac{8^n}{(3^2)^n} = 1 + \frac{8}{9} + \frac{64}{81} + \cdots = \frac{1}{1-\frac{8}{9}} = \frac{1}{\frac{1}{9}} = 9$

(f) $\displaystyle\sum_{n=0}^{\infty} \frac{2^n}{5^n}$ $\quad a = 1 \quad r = \frac{2}{5} < 1 \; C$ to $\frac{1}{1-\frac{2}{5}} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$

(g) $\displaystyle\sum_{n=0}^{\infty} \frac{1+2^n}{5^n}$ $= \displaystyle\sum_{n=0}^{\infty} \frac{1}{5^n} + \sum_{n=0}^{\infty} \frac{2^n}{5^n} = \frac{5}{4} + \frac{5}{3} = \frac{15+20}{12} = \frac{35}{12}$

$a = 1 \quad r = \frac{1}{5}$

$\text{sum} = \frac{1}{1-\frac{1}{5}} = \frac{5}{4}$

$a = 1 \quad r = \frac{2}{5}$

$\text{sum} = \frac{1}{1-\frac{2}{5}} = \frac{5}{3}$

3 The Harmonic Series

The harmonic series is

$$\sum_{n=1} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

It diverges.

Proof:

Does this series converge or diverge?

$$\cancel{\sum_{n=1} \frac{1}{n(n+1)}}$$

$$S_1 = 1$$

$$S_2 = 1 + \frac{1}{2}$$

$$S_4 = \underbrace{1 + \frac{1}{2}}_{S_2} + \left(\frac{1}{3} + \frac{1}{4}\right) = S_2 + \left(\frac{1}{3} + \frac{1}{4}\right) > S_2 + \left(\frac{1}{4} + \frac{1}{4}\right) = S_2 + \frac{1}{2}$$

$$S_8 = \underbrace{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}}_{S_4} + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > S_4 + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$

$$= S_4 + \frac{1}{2}$$

8 of these

$$S_{16} = \underbrace{1 + \frac{1}{2} + \cdots + \frac{1}{8}}_{S_8} + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) > S_8 + \left(\frac{1}{16} + \frac{1}{16} + \cdots + \frac{1}{16}\right)$$

$$= S_8 + \frac{1}{2}$$

11.3 p-series and the integral test

1 p-Series

A p-series has the form

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

1.0.1 Example:

$p = \frac{1}{2}$: $\frac{1}{1} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{2} + \frac{1}{\sqrt{5}} + \cdots$ Diverges

$p = 1$: $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$ Diverges

$p = 2$: $\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots$ Converge

$p = 3$: $1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \frac{1}{125} + \cdots$ Converge

1.1 The p-series with $p = 1$

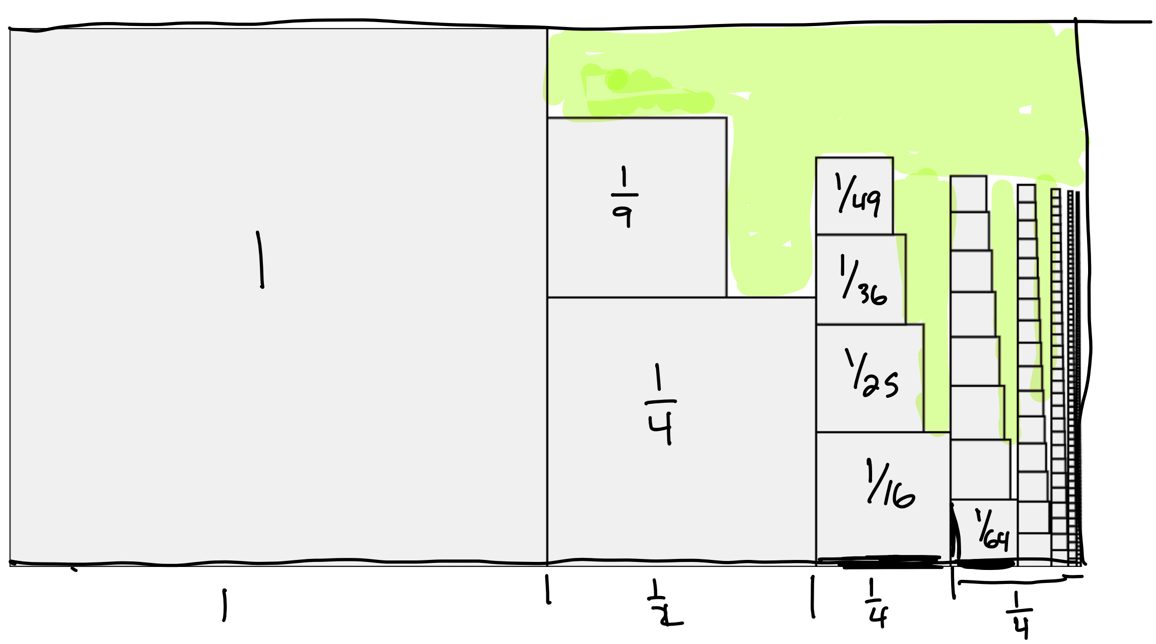
This is the harmonic series. We know it diverges.

1.2 The p-series with $p = 2$

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots$$

Does it converge?



	A	B	C	D	E	F	G	H	I	J	K
1	i	1/i^2	sum		i	1/i^2	sum		i	1/i^2	sum
2	1	1	1		11	0.008264463	1.5580322		1000	0.000001	1.64393457
3	2	0.2500000	1.2500000		12	0.0069444	1.5649766		2000	0.00000025	1.644434195
4	3	0.1111111	1.3611111		13	0.0059172	1.5708938		3000	0.000000111	1.644600792
5	4	0.0625000	1.4236111		14	0.0051020	1.5759958		4000	0.000000062	1.644684101
6	5	0.0400000	1.4636111		15	0.0044444	1.5804403		5000	0.00000004	1.64473409
7	6	0.0277778	1.4913889		16	0.0039063	1.5843465		6000	0.000000028	1.644767417
8	7	0.0204082	1.5117971		17	0.0034602	1.5878067		7000	0.00000002	1.644791223
9	8	0.0156250	1.5274221		18	0.0030864	1.5908932		8000	0.000000016	1.644809078
10	9	0.0123457	1.5397677		19	0.0027701	1.5936632		9000	0.000000012	1.644822965
11	10	0.0100000	1.5497677		20	0.0025000	1.5961632		10000	0.00000001	1.644834075

The exact sum of this series eluded mathematician for years. This was called the Basel Problem, and was solved by Euler in 1734. He found that the sum is . . . $\frac{\pi^2}{6}$

3Blue1Brown has a great video:

1.3 Convergence or divergence

***p*-Series** The p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

- converges if $p > 1$
- diverges if $p \leq 1$
- is the harmonic series when $p = 1$
- has no simple formula for its sum.

C code to generate the third column of partial sums shown above.

```
#include <stdio.h>
int main(){
  double sum = 0;
  double term = 0;
  for (int i=1; i<10001; i++){
   term = 1/(float) i/ (float) i;
   sum += term;
   if (!(i%1000)) printf("%d  %.9f %.9lf \n",i, term, sum);
  }
return 0;
}
```

1.3.1 Example:

$1 \cdots + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \cdots$ D

Do these series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{3}{n^2} = 3\sum \frac{1}{n^2}$$

C

$p = 2$

$$\sum_{n=1}^{\infty} \frac{n^3+n}{n^4}$$ D

$$\sum \frac{n^3}{n^4} + \sum \frac{n}{n^4}$$

$$\sum \frac{1}{n} + \sum \frac{1}{n^3}$$

D C

$$\sum_{n=1}^{\infty} \frac{1}{n+10}$$ D

It's the harmonic series with the first 10 terms missing

$$\sum_{n=1}^{\infty} \frac{1}{n^3+n} < \sum \frac{1}{n^3}$$

C C

$$\boxed{\Sigma \frac{1}{n^3+n} < \Sigma \frac{1}{n^3} = \text{Finite}}$$

2 The Integral Test

Integral Test

Let $f(x)$ be a continuous function that decreases to 0 as $x \to \infty$, and Let $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n \text{ is convergent iff } \int_1^{\infty} f(x)\,dx \text{ is convergent.}$$

Note: This test says that $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x)\,dx$ do the same thing. It does NOT say that they are equal.

2.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{\ln n}{n}$$